



ARTIFICIAL INTELLIGENCE AND ITS ROLE IN CYBERSECURITY

An Overview



by Christina Banks

CYB-670 CAPSTONE PAPER - UNIVERSITY OF MARYLAND GLOBAL CAMPUS

Instructor: Dr. Jose Da Cruz

Introduction

Artificial intelligence (AI) has rapidly transformed the technological landscape, with various industries adopting its usage and redefining the way humans interact with machines. AI is generally thought of as the ability of computers to solve problems in a manner similar to human cognition but with limited intervention from humans (NASA, n.d.). Artificial intelligence consists of machine learning (ML), natural language processing, neural networks, computer vision and deep learning, allowing systems to analyze data, identify patterns, and automate processes with remarkable efficiency. The use of AI has enhanced productivity and optimized decision-making in the public sector, healthcare, transportation, and cybersecurity, to name a few.

This paper not only explores the fundamentals of artificial intelligence and its widespread adoptions but, specifically, its role in strengthening cybersecurity measures. The examination of AI-powered threat detection, automated responses, and predictive analytics highlights the significance of AI in fortifying digital security in an increasingly connected world. AI can assist organizations in detecting, preventing and responding to security threats in real time, significantly improving organizations' security posture.

AI Across Sectors and in Cybersecurity - Use Cases

In recent years, the public sector has become more receptive to using AI to harness its power for the sake of enhancing service delivery, optimizing operations and citizen engagement, and bolstering security measures. The in-house development and deployment of chatbots and virtual assistants allows agencies to provide timely, relevant and accurate information to users, by utilizing natural language processing technologies to understand and interpret user queries effectively. In the interest of public health, AI is being used for disease surveillance, analyzing large datasets to better predict and detect disease outbreaks and, in turn, assisting governments in the development of better containment strategies (Sethi, 2025).

In healthcare, more broadly, the use of AI is increasingly being explored as a means of improving the speed and accuracy of medical diagnoses, tailoring treatment plans, and streamlining administrative and operational processes in hospitals and clinics to raise the level of patient care and outcomes (Alowais *et. al.*, 2023.) Currently, AI is used predominantly for patient scheduling and to move patients through the check-in and intake processes more quickly via wearable devices, equipped with GPS that contain patient-specific information. For patients that may not speak the same language as their doctor, tools such as IBM Watson Text to Speech can help medical professionals bridge that gap and expand healthcare accessibility (Laviola, 2024.)

Another application of AI and, more specifically, computer vision and machine learning, is traffic control and infrastructure improvements. Computer vision is a subset of AI “that uses machine learning and neural networks to teach computers and systems to derive meaningful information from digital images, videos and other visual inputs—and to make recommendations or take actions when they see defects or issues.” (IBM, 2021.) Cameras set up at intersections provide the visual data necessary for machine learning algorithms to analyze car and pedestrian density, the flow of traffic, and traffic light timing. Using this data, ML algorithms learn, autonomously and in real time, which intersections are high risk and can develop insights for traffic safety improvements and efficiency (Sakhuja, 2023.)

One of the most important applications of AI has been in cybersecurity. As cyber threats grow in complexity and frequency, traditional security measures can struggle to keep pace. AI models can be trained to recognize malware and malware-related signatures. Over time, AI learns how to recognize emerging threats, including new malware strains. With the ability to analyze vast amounts of network traffic, AI can also enhance threat intelligence by learning to recognize patterns of intrusion and other anomalous behaviors and predict the likelihood of a successful attack (Buczak & Guven, 2016.) Additionally, AI-powered tools speed up incident response, containing damage due to a breach (Goodman, 2025).

Strengths, Weaknesses, Opportunities and Threats (SWOT) for using AI

SWOT, or strengths, weaknesses, opportunities and threats, is a strategic planning tool used for identifying and analyzing the strengths, weaknesses, opportunities and threats related to a business or project (Bigelow, 2023.) The purpose is to inform decision makers of the internal and external factors that can affect, for better or worse, the outcomes of those chosen business decisions.

Strengths:

Understanding strengths help organizations maximize their potential and maintain a competitive edge in the market. AI is a fantastic, scalable tool for performing mundane, repetitive, and tedious tasks. It can sort through and analyze large amounts of data, freeing humans to perform more complex tasks, ultimately, saving time and money (Makos, 2024.)

In cybersecurity, AI is used for 24/7 security and compliance monitoring, and incident response, alongside traditional methods (Kerwin, 2024). AI is frequently used to parse large amounts of log data to identify anomalies in network traffic and block malicious traffic automatically. Historical data is used to train AI in pattern recognition, resulting in predictive insights that allow AI to recognize potential vulnerabilities in a network (Sufi, 2023.) Paired with traditional cyber threat intelligence gathering methods, organizations are

better positioned to fortify their defenses and minimize their attack surface. However, this convenience and efficiency comes with some serious considerations.

Weaknesses:

AI models that rely on biased or insufficient data can result in biased outcomes or hinder well-reasoned decision making. Algorithmic bias is a major concern in that it can reinforce and even exacerbate existing inequalities in society. In a healthcare context, the lack of diverse datasets, representing different classes of people, can lead to delayed response in clinical decision making (Jain, *et. al.*, 2023.) For example, AI trained to recognize a particular disease as having slow progression without considering variances in disease progression between ethnic or gender groups, can lead to poorer clinical outcomes (Norori, *et. al.*, 2021.)

Tying into that, AI does not grasp context and nuance well and will sometimes completely misconstrue or misrepresent data to fill contextual gaps, so a human is still necessary, in many cases, to review an AI model for quality assurance in relation to a particular application (Ghosh, 2024).

Where cybersecurity is concerned, humans must continuously stay up to date on cyber threats and methods and regularly audit and re-train AI models to prevent AI drift. AI drift is a phenomenon that occurs when data that the model was trained on is out-of-date or no longer relevant and, as a result, produces unwanted or unexpected results such as false

positives; this, in turn, results in much weaker defenses and in some cases creates new vulnerabilities (Tipton, 2025.) Another weakness is the cost to implement. AI requires vast amounts of storage, computing power, and professional expertise, which adds to a company's overhead. Smaller businesses will have a harder time keeping up with these costs.

Opportunities:

Identifying opportunities allows businesses to strategize effectively and capitalize on favorable conditions in the external environment. As costs associated with implementing AI continue to decline, these tools become more accessible to smaller businesses, granting them access to automated security solutions. The integration of AI with other traditional cybersecurity research, monitoring, mitigation, and remediation tools, at a lower cost, provides businesses with the opportunity to better shore up their defenses.

Threats:

Threats are also defined as external factors that could cause problems for the organization. An adversary can possibly use these same tools more successfully in an attack than the business can, defensively. AI can also be used by adversaries to engineer more advanced malware that can evade current defense measures used by enterprises (Brundage, *et. al.*, 2018.)

Data privacy poses another major ethical and legal threat because training AI relies on a model's access to immense amounts of data, some of which can be sensitive and attributable to individuals, also known as personally identifiable information or PII (Makos, 2024.) The use of AI in monitoring can lead to potential privacy violations if not properly managed. Hackers could steal this data and use it as leverage in ransomware attacks. Hackers could also poison AI training data by breaching the database(s) where the data is stored. As a result, the AI's ability to correctly recognize and respond to attacks and vulnerabilities is destroyed (Eggers & Sample, 2020, pg #6; Watkins, 2024).

Specific Recommendations for Applying AI RMF Framework and its Associated Playbook

The Artificial Intelligence Risk Management Framework (AI RMF) is a guidance document developed by the National Institute of Standards and Technology to help organizations manage risks associated with AI systems. It aims to enhance the trustworthiness of AI technologies by addressing aspects such as fairness, accountability, and transparency through the AI lifecycle (NIST, 2023.) The AI lifecycle is the process by which a business problem is discovered and then an AI solution is identified and developed through the three steps of the AI lifecycle: design, development and deployment (GSA, n.d.). The four core functions integrated into the AI lifecycle and that make up the AI RMF are govern, map, measure, and manage.

AI governance involves following established regulatory requirements to prevent discrimination and similar harm, and defining clear policies, roles, and responsibilities to ensure a culture of AI risk management within the organization (The White House, 2023). Organizations should assign accountability by, for example, having dedicated AI teams according to the AI RMF core functions. Any business in any industry that wants to adopt AI tools should engage with diverse stakeholders from various departments within the org in the AI development and deployment processes (Vanta, 2025). The framework should also be tailored to the organization's specific objectives, risk appetite, and regulatory environment, and vulnerabilities should be studied and documented (Siddiqui, n.d.) The playbook itself is a great way to align an organization's AI implementation with the AI RMF as it offers suggested actions, references, and guidance aligned with the framework.

Summary

This paper only scratches the surface of what is becoming a deep and multifaceted technology and discipline. AI has evolved into a transformative force across multiple industries, including government, healthcare, finance, manufacturing, and especially cybersecurity. Governments leverage AI for public services, defense, and intelligence analysis, improving decision-making and operational efficiency. In the private sector, AI enhances customer experience, optimizes supply chains, and automates complex processes. Organizations use AI to strengthen their defenses against cyber threats such as phishing, ransomware, and advanced persistent threats (APT), improving resilience in an increasingly digital world.

A comprehensive SWOT analysis of AI in a cybersecurity context reveals its strengths, weaknesses, opportunities, and threats. AI has great potential, however, weaknesses such as bias in AI models, lack of transparency, and high implementation costs pose challenges. Despite this, innovative opportunities abound in improved fraud detection, enhanced predictive analytics, and autonomous cybersecurity systems. Adversarial AI and data privacy concerns are just a couple of ever-present threats for organizations.

To mitigate risks involved in implementing AI, the NIST AI Risk Management Framework and accompanying playbook provide great guidance on managing AI-related risks and enhancing transparency. By applying AI RMF principles, organizations can identify potential biases, assess cybersecurity risks and implement robust governance practices. While AI offers significant benefits, it is crucial to address its inherent weaknesses and threats to fully harness its power and deploy it responsibly and securely.

References

- Alowais, S.A., Alghamdi, S.S., Alsuhebany, Alqahtani, T., Abdulrahman, A.I., Almohareb, S.N., Aldairem, A., Alrashed, M., Bin Saleh, K., Badreldin, H., Al Yami, M.S., Al Harbi, S. & Albekairy, A.M. . (2023 September 22). Revolutionizing healthcare: the role of artificial intelligence in clinical practice. *BMC Med Educ* **23**, 689.
<https://doi.org/10.1186/s12909-023-04698-z>
- Bigelow, S. J., Pratt, M.K., & Tucci, L. (2023 February). SWOT analysis (strengths, weaknesses, opportunities and threats analysis). *TechTarget*.
<https://www.techtarget.com/searchcio/definition/SWOT-analysis-strengths-weaknesses-opportunities-and-threats-analysis>
- Buczak, A.L. & Guven E. (2015 October 26). A Survey of Data Mining and Machine Learning Methods for Cyber Security Intrusion Detection. *IEEE Communications Surveys & Tutorials* 18(2), 1153-1176. <https://ieeexplore.ieee.org/document/7307098>
- Bundage, M., Shahar, A., Clark, J., Toner, H., Eckersley, P., Garfinkel, B., Dafoe, A., Scharre, P., Zeizoff, T., Filar, B., Anderson, H., Roff, H., Allen, G., Steinhardt, J., Flynn C., Helgeartaigh, S., Beard, S., Belfield, H., Farquhar, S.,...& Amodei, D. (2018 February).

The Malicious Use of Artificial Intelligence: Forecasting, Prevention, and Mitigation.

The Future of Humanity Institute. <https://arxiv.org/pdf/1802.07228>

DataGuard Insights. The growing data privacy concerns with AI: What you need to know.

DataGuard. <https://www.dataguard.com/blog/growing-data-privacy-concerns-ai/>

Eggers, E. & Sample C. (2020 December). Vulnerabilities in Artificial Intelligence and

Machine Learning Applications and Data. *Idaho National Laboratory*.

https://inldigitallibrary.inl.gov/sites/sti/sti/Sort_57369.pdf

General Services Administration. (n.d.). Understanding and managing the AI lifecycle

<https://coe.gsa.gov/coe/ai-guide-for-government/understanding-managing-ai-lifecycle/>

Ghosh, R. (2024 September 12). Strengths, Weaknesses, Opportunities, and Threats: A

Comprehensive SWOT Analysis of AI and Human Expertise in Peer Review. *The*

Scholarly Kitchen. [https://scholarlykitchen.sspnet.org/2024/09/12/strengths-](https://scholarlykitchen.sspnet.org/2024/09/12/strengths-weaknesses-opportunities-and-threats-a-comprehensive-swot-analysis-of-ai-and-human-expertise-in-peer-review/)

[weaknesses-opportunities-and-threats-a-comprehensive-swot-analysis-of-ai-and-human-expertise-in-peer-review/](https://scholarlykitchen.sspnet.org/2024/09/12/strengths-weaknesses-opportunities-and-threats-a-comprehensive-swot-analysis-of-ai-and-human-expertise-in-peer-review/)

Goodman, C. (2025 January 16). AI in Cybersecurity: Transforming Threat Detection and Prevention. Balbix. <https://www.balbix.com/insights/artificial-intelligence-in-cybersecurity/>

IBM. (2021 July 7). What is computer vision?. *IBM*.
<https://www.ibm.com/think/topics/computer-vision>

Jain, A., Brooks, J.R., Alford, C.C., Chang, C. S., Mueller, N.M., Umscheid, C., & Bierman, A.S. (2023 June 2). Awareness of Racial and Ethnic Bias and Potential Solutions to Address Bias With Use of Health Care Algorithms. *JAMA Health Forum*, 4(6).
<https://jamanetwork.com/journals/jama-health-forum/fullarticle/2805595>

Kerwin, J. (2024 August 1). What Is the Role of AI in Cybersecurity?. *Excelsior University*.
<https://www.excelsior.edu/article/ai-in-cybersecurity/>

Laviola, E. (2024 October 3). AI in Healthcare: How It's Used and Future Use Cases. *HealthTech Magazine*. <http://healthtechmagazine.net/article/2024/10/ai-in-healthcare-how-its-used-future-perfcon>

Makos, J. (2024 July 28). SWOT Analysis of Artificial Intelligence. *Pestle Analysis*.

<https://pestleanalysis.com/swot-analysis-of-artificial-intelligence/>

NASA. (n.d.). What is Artificial Intelligence?. *National Aeronautics Space Administration*.

<https://www.nasa.gov/what-is-artificial-intelligence/>

NIST (2025 January 7). NIST AI RMF: Everything you need to know. *Vanta*.

<https://www.vanta.com/resources/nist-ai-risk-management-framework>

Norori, N., Hu, Q., Aellen, F. M., Faraci, F.D. & Tzovara, A. (2021 October 8). Addressing bias in big data and AI for health care: A call for open science. *Patterns* 2(10).

<https://www.sciencedirect.com/science/article/pii/S2666389921002026>

Rashid, A.B. & Kausik, A.K. (2024 December). AI revolutionizing industries worldwide: A comprehensive overview of its diverse applications. *Hybrid Advances*.

<https://www.sciencedirect.com/science/article/pii/S2773207X24001386>

Sakhuja, A. (2023 December). Intelligent Traffic Management System using Computer Vision and Machine Learning. *Innovative Research Thoughts*. 9. 1-10.

10.36676/irt.2023-v9i5-001

Sethi, A.A. (2025 March 5). The Top AI Applications in Government Contributing to Public Sector Innovation. *HCLTech*. <https://www.hcltech.com/trends-and-insights/top-ai-applications-government-contributing-public-sector-innovation>

Siddiqui, M. (n.d.). Aligning with the NIST AI RMF Using a Step-by-Step Playbook. *CyberSaint Security*. <https://www.cybersaint.io/blog/nist-ai-rmf-playbook>

Sufi, F. (2023 December 9). A global cyber-threat intelligence system with artificial intelligence and convolutional neural network. *Decision Analytics Journal*, vol. 9. <https://www.sciencedirect.com/science/article/pii/S2772662223002047>

The White House. (2023). RECOMMENDATIONS: Implementing the NIST AI RMF with a Rights-Respecting Approach. *The National Artificial Intelligence Advisory Committee*. https://ai.gov/wp-content/uploads/2023/11/Recommendations_Implementing-the-NIST-AI-RMF-with-a-Rights-Respecting-Approach.pdf

Tipton, T. (2025 March 5). The Ghost in the Algorithm: Unintended Consequences of AI in Cybersecurity. *Arctiq*. <https://arctiq.com/blog/the-ghost-in-the-algorithm-unintended-consequences-of-ai-in-cybersecurity>

Watkins, O. (2024 April 19). 4 use cases for AI in cyber security. *Red Hat Blog*.

<https://www.redhat.com/en/blog/4-use-cases-ai-cyber-security>